

1. Docker Compose

Definition

Docker Compose is a tool used to define and run multiple Docker containers using a single YAML file (docker-compose.yml or compose.yml).

Instead of running many docker run commands, we define all services in one file and start them together.

Why Docker Compose?

Suppose you have a web application that needs:

Frontend

Backend

Database

Without Compose, you must start each container manually.

With Docker Compose, one command starts all containers.

Advantages

Manages multiple containers.

Easy networking between containers.

Volume management.

Easy to start and stop applications.

Suitable for development and testing.

Example

```
version: "3"
```

```
services:
```

```
  web:
```

```
    image: nginx
```

db:

image: mysql

Common Commands

docker compose up

docker compose down

docker compose ps

docker compose logs

docker compose restart

2. Drawbacks of Docker

Although Docker is very useful, it has some limitations.

1. Single Host Limitation

Docker manages containers on only one machine.

2. No Automatic Scaling

If traffic increases, Docker alone cannot automatically create more containers.

3. No Self-Healing

If a container crashes, Docker does not automatically recreate it.

4. Manual Load Balancing

Traffic distribution between containers must be managed manually.

5. Complex for Large Applications

Managing hundreds of containers manually becomes difficult.

6. Monitoring and Logging

Docker provides only basic logging. Large environments need additional tools.

3. What is Kubernetes?

Definition

Kubernetes (K8s) is an open-source container orchestration platform that automates deployment, scaling, networking, and management of containerized applications.

Why Kubernetes

Docker runs containers.

Kubernetes manages those containers across multiple servers.

Features

Automatic deployment

Auto scaling

Self-healing

Load balancing

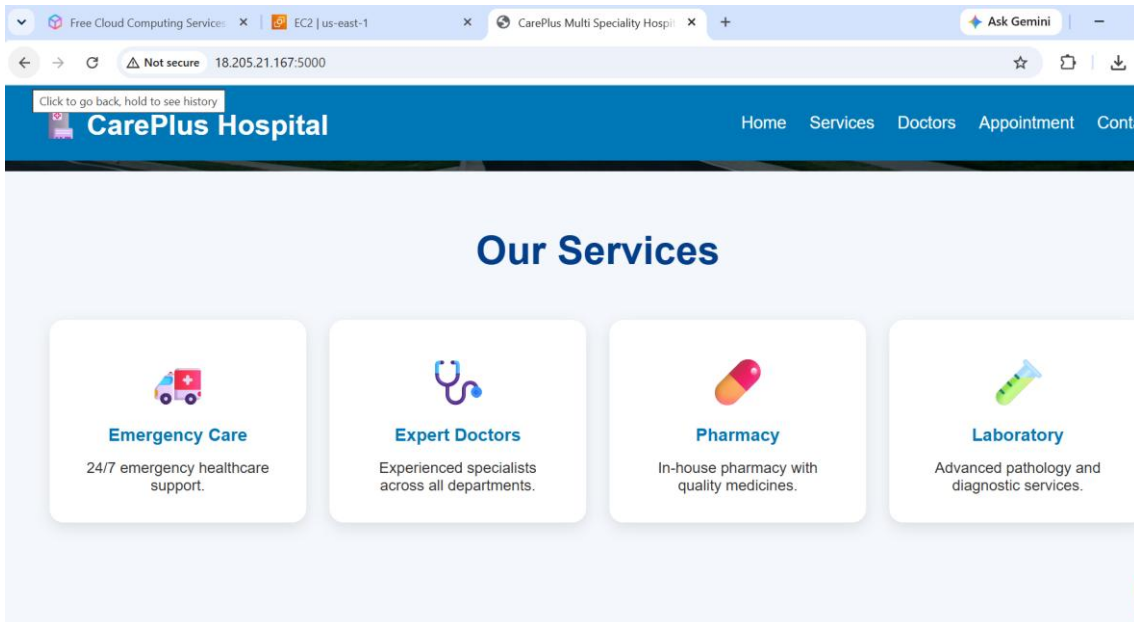
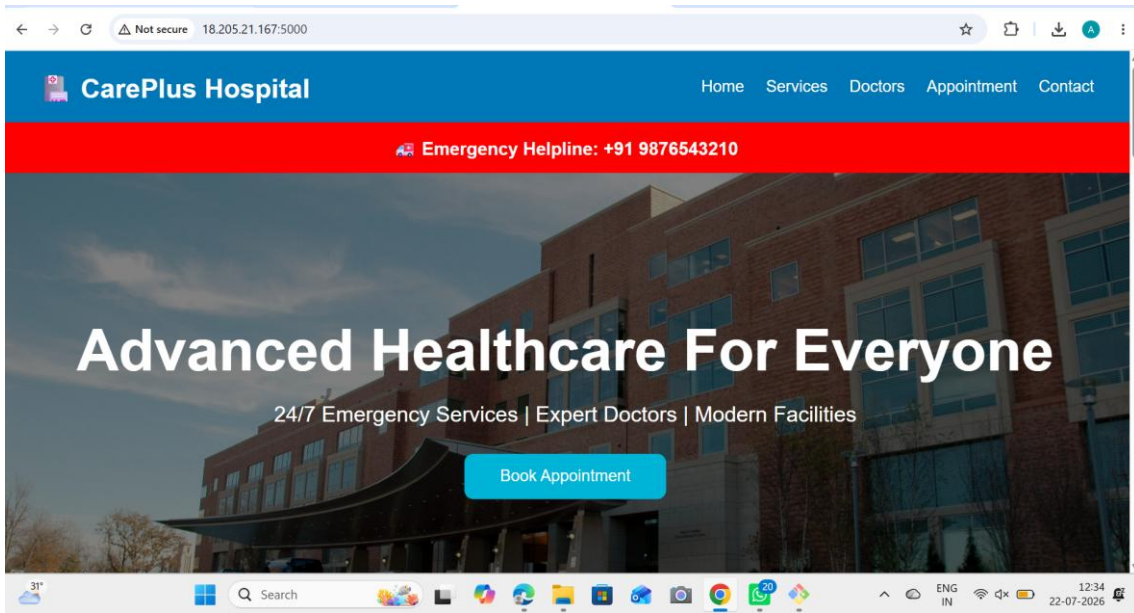
Rolling updates

Service discovery

Storage management

I created website using three files attached in server

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Neurologist



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Orthopedic Specialist

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```
root@ip-172-31-17-198: /home/ubuntu
=> [flask-app internal] load .dockerignore
=> transferring context: 2B
=> [flask-app internal] load build context
=> transferring context: 12.27kB
=> [flask-app build 1/4] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f75559aabb1acd1e13cf1731b1b
=> resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f75559aabb1acd1e13cf1731b1b
=> sha256:ea56f685404adf81680322f152d2cfec2115b30dda481c2c450078315beb508 251B / 251B
=> sha256:b2ec99836ae8c0332e98514e4eca80391df4e81a9daa075048c2e3df3801d 1.29MB / 1.29MB
=> sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d10c2c83d710affa286c08 13.88MB / 13.88MB
=> sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 29.78MB / 29.78MB
=> extracting sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d
=> extracting sha256:b36c38b36ae8c0332e98514e4eca80391df4e81a9daa075048c2e3df3801d
=> extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d10c2c83d710affa286c08
=> extracting sha256:ea56f685404adf81680322f152d2cfec2115b30dda481c2c450078315beb508
=> [flask-app build 2/4] WORKDIR /app
=> [flask-app build 3/4] COPY app.py .
=> [flask-app build 4/4] RUN pip install --no-cache-dir flask
=> [flask-app stage-1 3/4] COPY --from=build /usr/local/lib/python3.9 /usr/local/lib/python3.9
=> [flask-app stage-1 4/4] COPY --from=build /app /app
=> [flask-app] exporting to image
=> exporting layers
=> exporting manifest sha256:60e65789fad67f26c7dfd3d4d457b8953f519c0b01919c0bcb32766baa359790
=> exporting config sha256:94a4fea490474c3e5365391dc564bfae28aaa72e8c1c04f092191aee12fd298
=> exporting attestation manifest sha256:f83c25e6164e6cf766cf4c08c25ee16624781aaab838d57a277ae673e780e4c
=> exporting manifest list sha256:a1d643d48369224051a6c326a3cf6c054f753f56090da41aaz7f0c7e5d31761e34
=> naming to docker.io/library/ubuntu-flask-app:latest
=> unpacking to docker.io/library/ubuntu-flask-app:latest
=> [flask-app] resolving provenance for metadata file
[+] Running 4/4
  flask-app      Built
  Network ubuntu_default Created
  Volume ubuntu_flask-volume Created
  Container flask-container Created
Attaching to flask-container
flask-container | * Serving Flask app 'app'
flask-container | * Debug mode: off
flask-container | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
flask-container | * Running on all addresses (0.0.0.0)
flask-container | * Running on http://172.0.0.1:5000
flask-container | * Running on http://172.18.0.2:5000
flask-container | Press CTRL+C to quit
flask-container | 157.51.56.51 - - [22/Jul/2026 07:03:51] "GET / HTTP/1.1" 200 -
flask-container | 157.51.56.51 - - [22/Jul/2026 07:03:58] "GET /favicon.ico HTTP/1.1" 404 -
Gracefully Stopping... press Ctrl+C again to force
Container flask-container Stopping
Container flask-container Stopped
flask-container exited with code 137
root@ip-172-31-17-198: /home/ubuntu#
```